

RISC-V Processor Core on FPGA

A fetch–decode–execute core in VHDL executing a RISC-V instruction subset, verified in simulation and on real hardware.

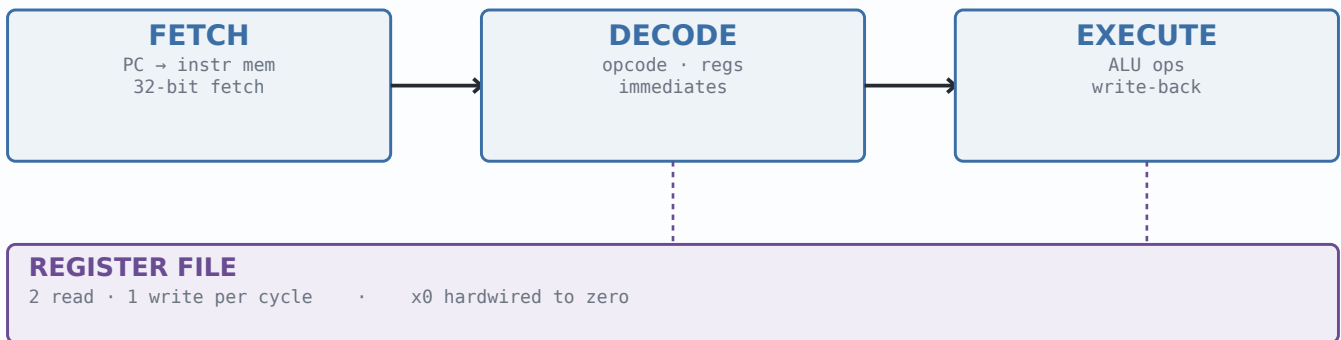


Fig. 1 – Classical fetch / decode / execute datapath with multi-port register file (x0 = 0).

Overview

This project designed and implemented a simple processor core in VHDL, deployed to an FPGA development board and capable of executing a subset of the open RISC-V instruction set architecture. The work covered the principal elements of processor microarchitecture — instruction fetch, decode and execute, a general-purpose register file, and an arithmetic logic unit — verified through both simulation and execution of assembly test programs.

Architecture & design

RISC-V was chosen for its clean, open and well-documented specification, an ideal vehicle for studying processor design. The core implements the classical pipeline structure:

FETCH	A program counter addresses instruction memory and retrieves the next 32-bit instruction word, with logic for sequential advance and control-flow redirection.
DECODE	Combinational decode logic extracts the opcode, source/destination register fields and immediate values from each RISC-V instruction format, generating the control signals that steer the datapath.
EXECUTE	The ALU performs the arithmetic and logical operations defined by the supported instruction subset, with results written back to the register file.
REGISTER FILE	A multi-port register bank implementing the RISC-V general-purpose registers — two simultaneous read ports and one write port per cycle, with x0 hardwired to zero per the spec.

Verification

Verification proceeded in two complementary stages. First, RTL simulation ran throughout development: testbenches drove the design in EDA Playground and waveforms were inspected to confirm correct instruction sequencing, register-file behaviour and ALU results cycle-by-cycle. Second, hand-written RISC-V assembly test programs were assembled and loaded into instruction memory, exercising each supported instruction and verifying architecturally correct end-state in the registers. The design was then synthesised and deployed to the FPGA board, demonstrating the core operating in real hardware.

Outcomes & skills demonstrated

The project produced a functioning soft-core processor and a concrete, implementation-level understanding of how an instruction set architecture maps onto digital hardware. It demonstrates VHDL design at register-transfer level, digital architecture (datapath and control design), the RISC-V ISA, testbench-driven verification and waveform analysis, and the FPGA synthesis and deployment flow.